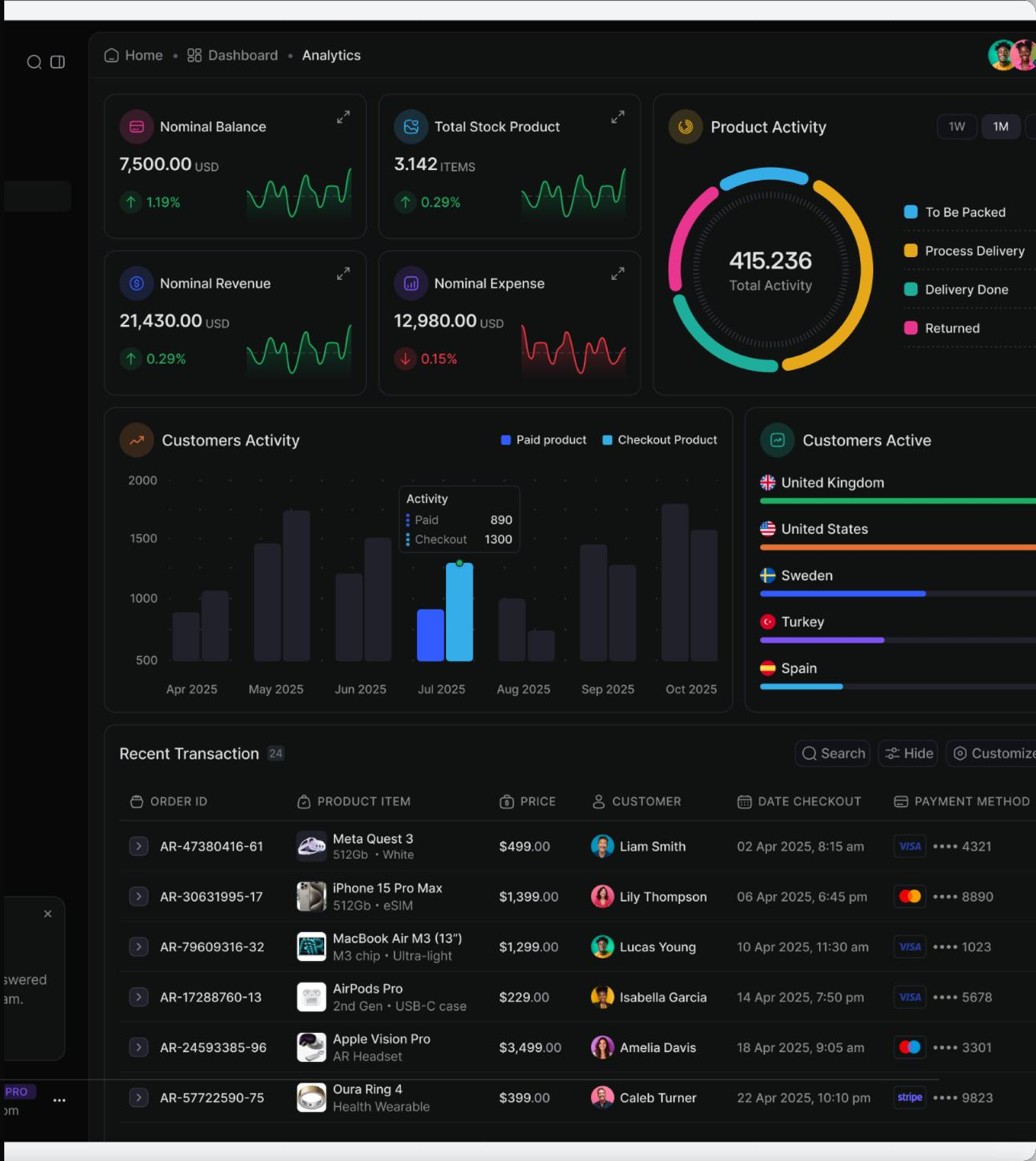
Modular Logistic Regression for PD

A scalable approach to Probability of Default modeling across distinct loan portfolios

The Challenge of Heterogeneity

- Monolithic models struggle with diverse risk profiles.
- Modular approaches isolate universal trends efficiently.
- Independently evaluates drivers unique to specific portfolios.
- Dramatically improves predictive accuracy and robustness.



Architectural Overview

The Common Module

- Applies globally across all loans.
- Sets baseline risk metrics.

Portfolio A Module

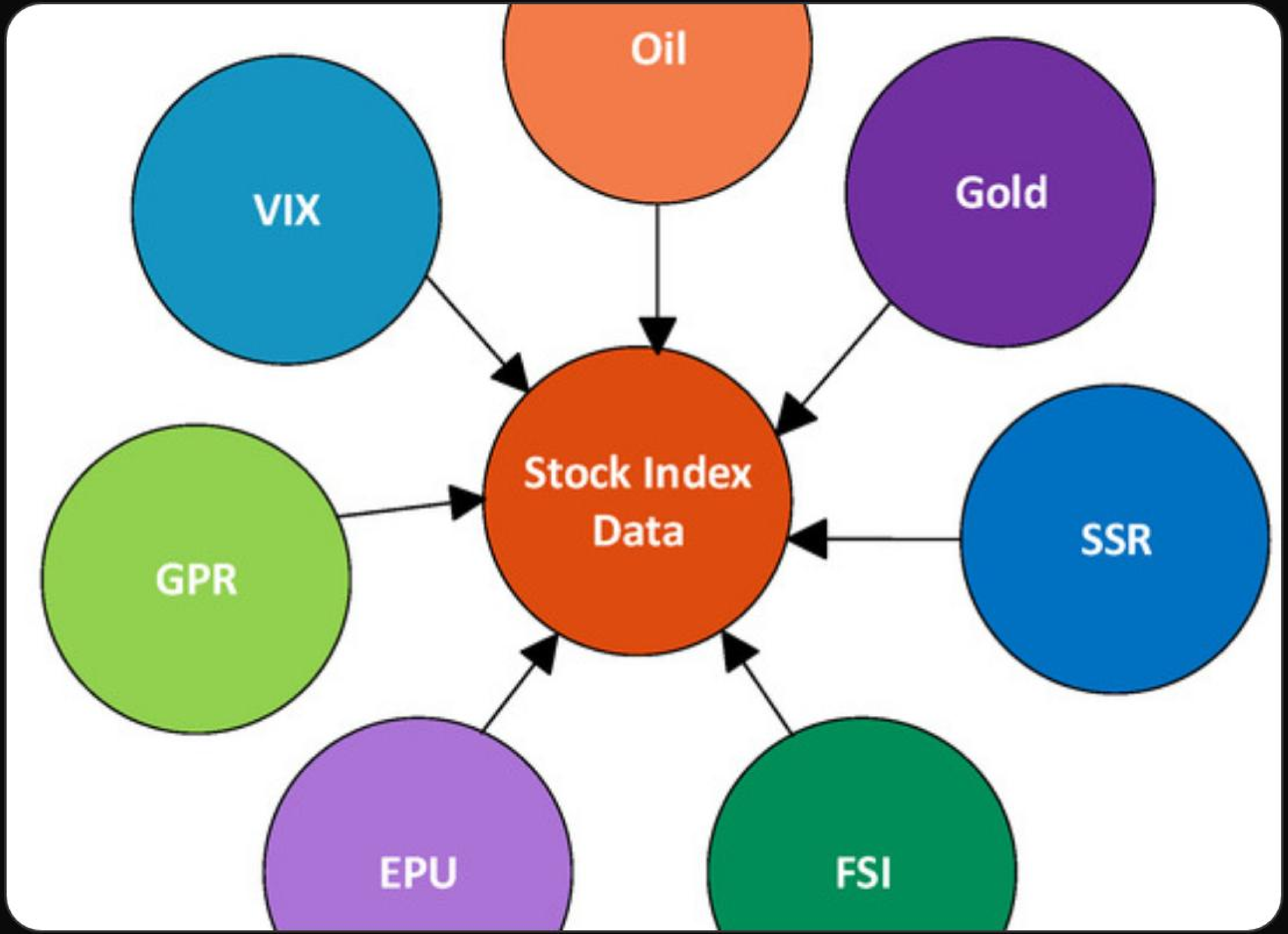
- Specialized logic tailored specifically for Portfolio A.
- Isolates unique risk drivers.

Portfolio B Module

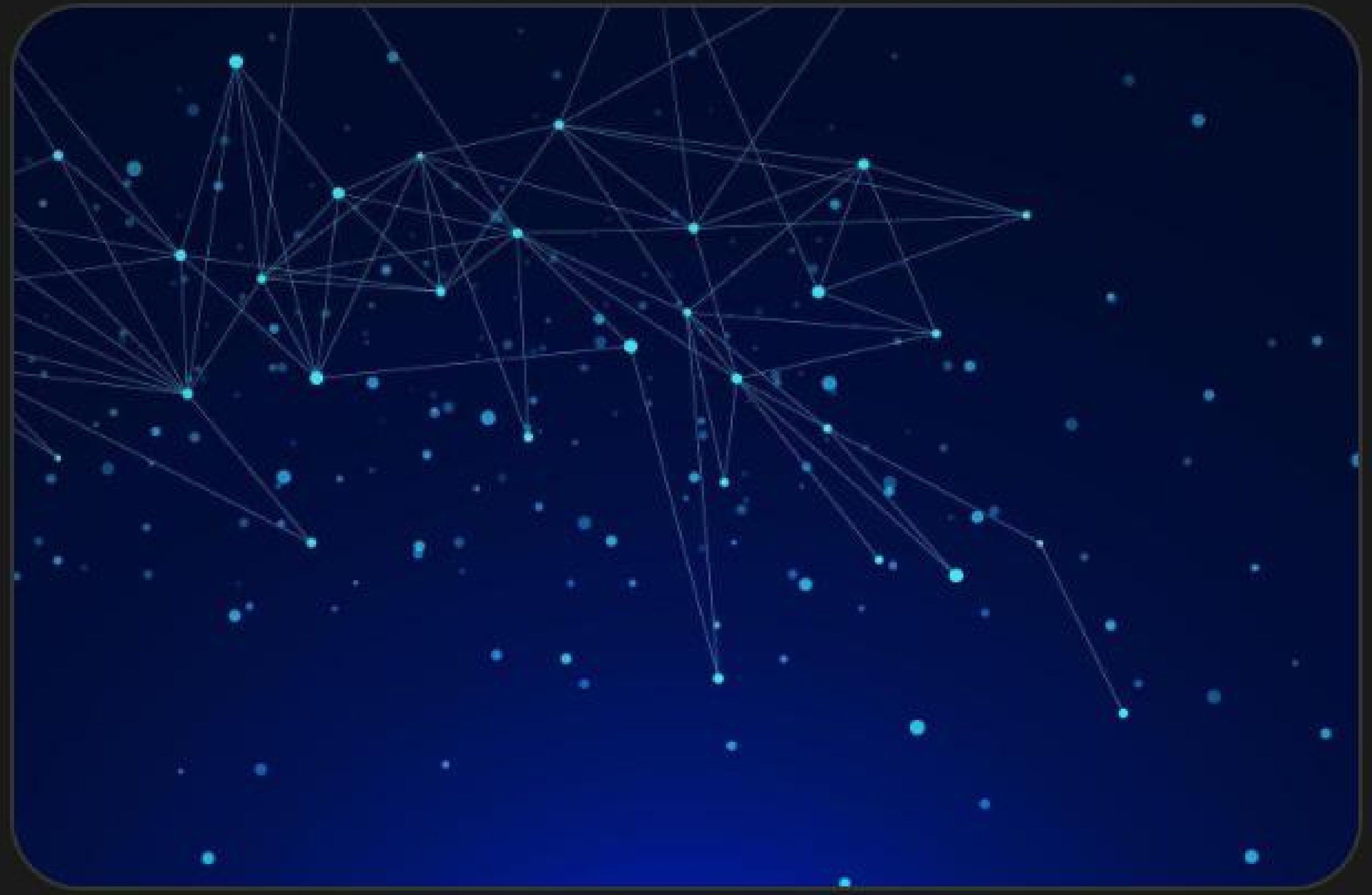
- Specialized logic tailored specifically for Portfolio B.
- Isolates unique risk drivers.

The Common Module

- Applies equally to all loans across diverse portfolios.
- Establishes baseline risk using universal metrics.
- May leverage standard macroeconomic indicators (e.g., GDP, inflation).
- Captures systemic risk, isolating idiosyncratic factors.

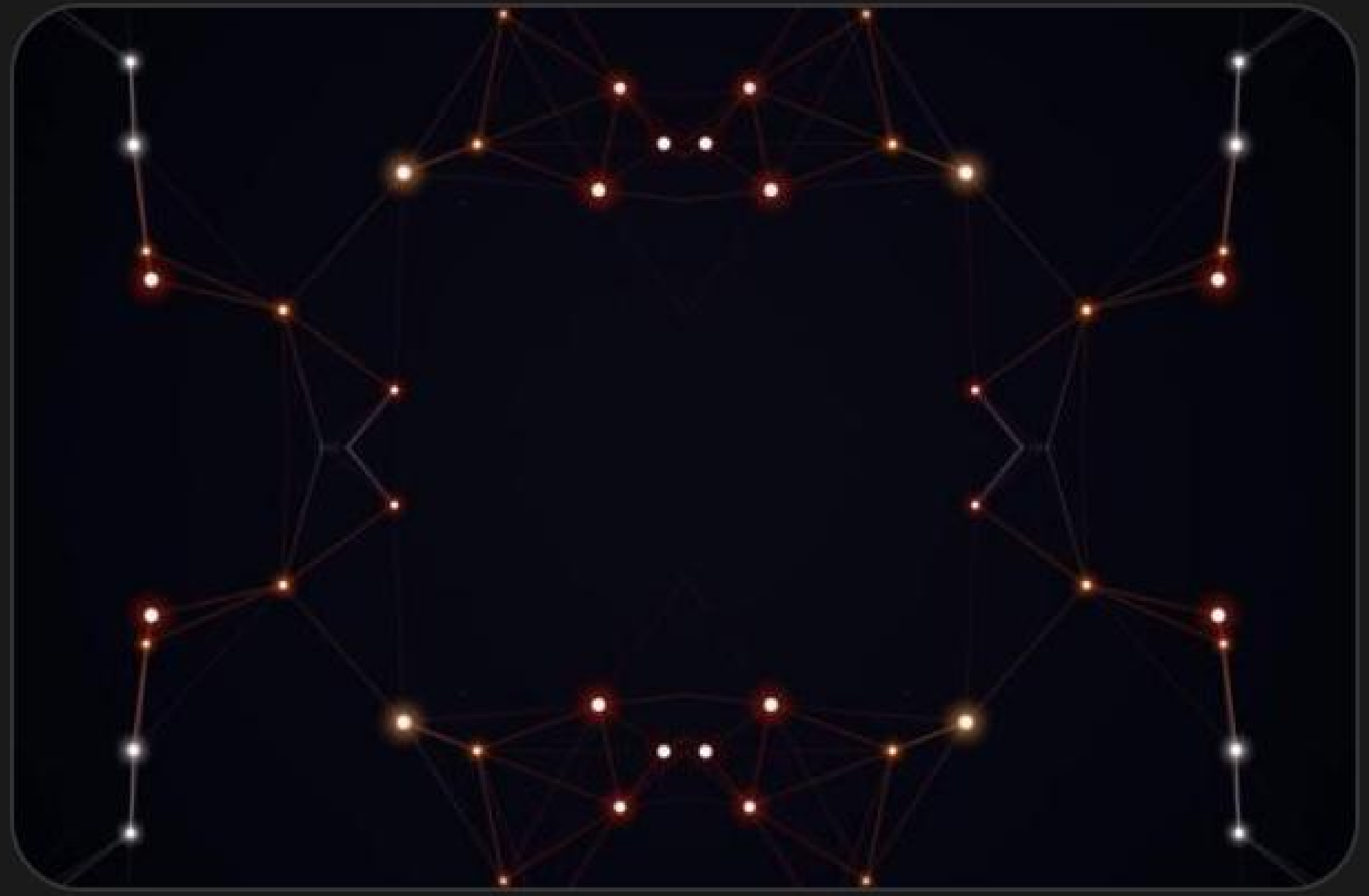


Portfolio-Specific Modules



Portfolio A

- Evaluates granular data specific to A.
- Measures idiosyncratic risk factors.



Portfolio B

- Evaluates specialized metrics for B.
- Assesses unique behavioral data.

Dynamic Integration

- Routing engine activates modules based on loan origin.

Mathematical Framework

Combining universal baseline risk with localized log-odds.

The Modular Equation

The Logistic Equation

$$P(\text{Default}) = \frac{1}{1 + e^{-(\alpha + \beta_c X_c + \beta_p X_p)}}$$

Variable Definitions

α : Global Base Log-Odds

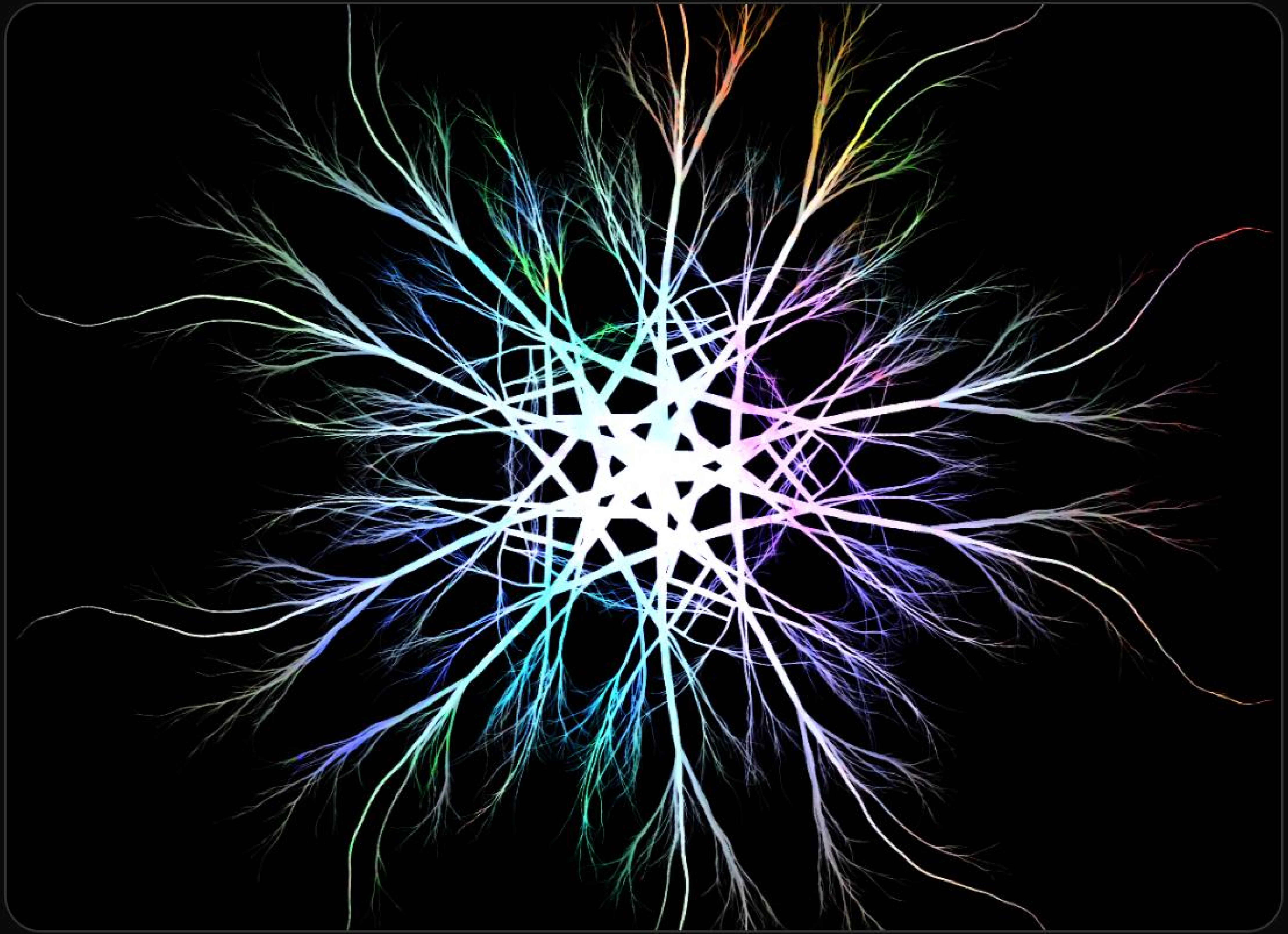
$\beta_c X_c$: Common Module Features

$\beta_p X_p$: Portfolio-Specific Features

The specific component β_p is conditionally activated via a binary flag depending on the loan's originating portfolio.

Implementation Stack

- Entire modular modeling framework built natively in **Python**.
- **pandas** powers the data orchestration, alignment, and robust feature engineering.
- **statsmodels** drives the logistic regression estimation and rigorous statistical diagnostics.
- Architecture ensures a highly scalable, reproducible, and production-ready analytical pipeline.



Performance Uplift (Gini Coefficient)

Image Sources

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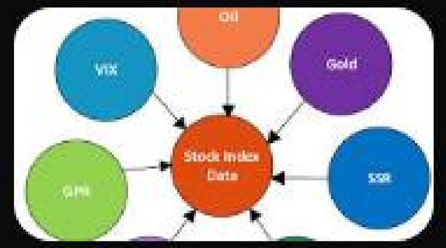

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